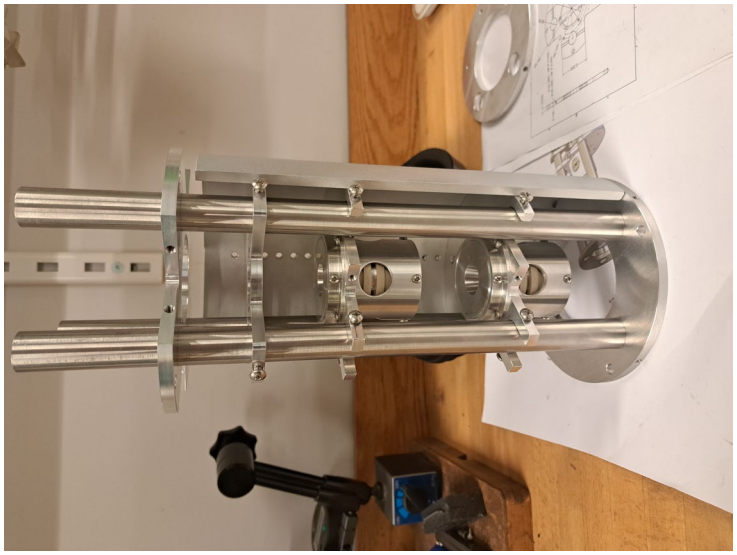
Time resolved measurement of electric currents in the range of 1 to 10 µA.

Supervisors: Linus, Philip

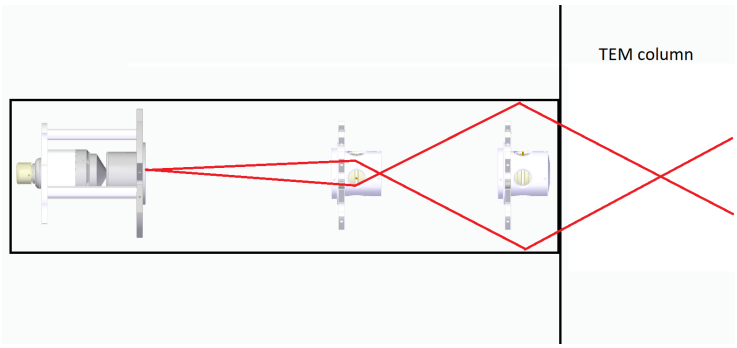
Background:

We are designing and building a prototype of a special purpose electron gun (figure 1) which can be used to enhance contrast in a transmission electron microscope based on the principle of a space charge phase plate (1).
To characterize the electron beam produced by this gun we plan to use a Faraday cup with a mechanical beam blank coupled to a time resolved readout circuit. (figure 2)

Two related student projects in last year's course were carried out. The one was a simulation of the heating behaviour of such a Faraday cup when irradiated by an electron beam, the other a simulation of the beam formation in the electron gun. Both were done in Comsol. Details can be obtained from your supervisors.

a 

b 



c

d 

Figure 1: a) The inner workings of the gun, b) The vacuum housing with pumps and pirani gauge, c) A drawing of the gun with 2 lenses and the thermionic source (on the left) attached to the TEM, d) thermionic electron source from an SEM.

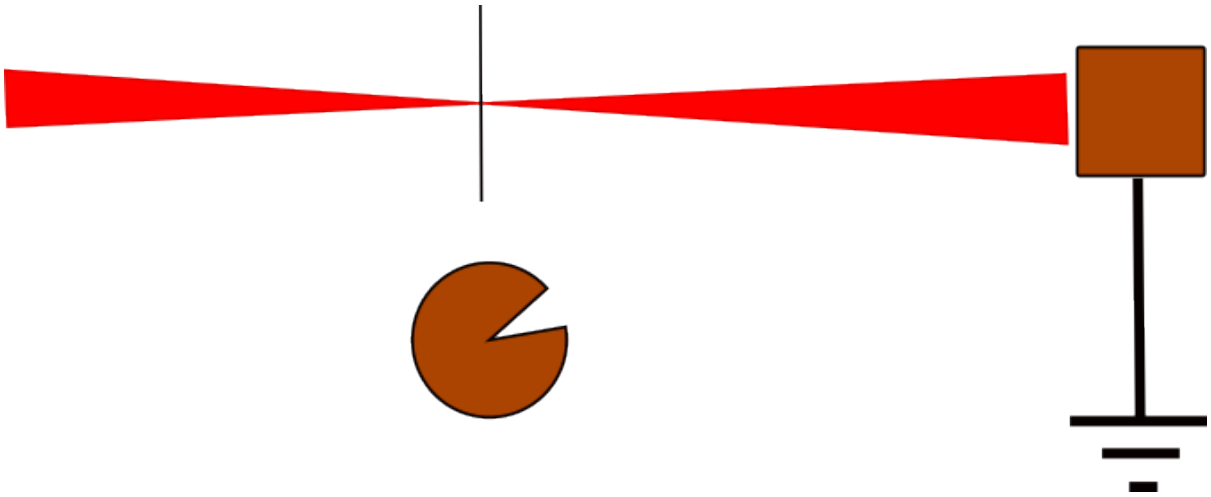


Figure 2: Measuring the beam current and diameter with a faraday cup and a rotating beam blank.

Project goal:

As a first step we want to design the readout circuit for the current from the Faraday cup to ground and test it with input signals produced by a signal generator.

References:

(1) https://www.researchgate.net/publication/331036561_Design_of_a_charged_particle_beam_phase_plate_for_transmission_electron_microscopy